

R V College of Engineering

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

II Semester B.E Regular/Supplementary Examinations July-2025.

(Common to AI, BT, CS, CD, CY, & IS)

Course : Number Theory, Vector Calculus And Computational Methods-MA221TC

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all the questions from Part A.
Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and four, 5 and 6, 7 and 8, 9 and 10.
3. Handbook of Mathematics is permitted.

Part A

Question No	Question	M	CO	BT
1.1	The remainder obtained when $(1! + 2! + 3! + \dots + 100!)$ is divided by 12 is_____.	02	2	2
1.2	If $693 = 3^x 7^y 11^z$, where x, y, z are positive integers, then the product of x, y, z is_____.	02	2	2
1.3	If the vector $\vec{F} = (2x^2y^2 + z^2)\hat{i} + (3xy^3 - x^2z)\hat{j} + (\lambda xy^2z + xy)\hat{k}$ is solenoidal, then the value of the constant λ is _____.	02	1	1
1.4	If $\vec{F} = (x + y + z)\hat{i} + (x + 2y - z)\hat{j} + (x + cy + 2z)\hat{k}$ is an irrotational vector field, then the value of $c =$ _____.	02	2	1
1.5	If $\text{curl } \vec{F} = 0$, then for any closed curve $C \int_C \vec{F} \cdot d\vec{r} =$ _____.	02	1	1
1.6	By Gauss divergence theorem , find $\iint_S (ax \hat{i} + by \hat{j} + cz \hat{k}) \cdot \hat{n} dS$ where S is the surface of the sphere $x^2 + y^2 + z^2 = 1$.	02	2	2
1.7	If $2 + \sqrt{3}$ and $2 - \sqrt{3}$ are the roots of an auxiliary equation then the corresponding differential equation is _____.	02	1	1
1.8	Solve $\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 0$.	02	1	1
1.9	Given $y = 2x^3 - 4x + 1$ for $x = -1, 0, 1$ then $\Delta^2\{y_0\} =$ _____.	02	1	1
1.10	Construct the forward difference table for the polynomial $y = x^2 - 3x + 1$ where $x = -1, 0, 1$.	02	2	2

Part B

Question No	Question	M	CO	BT
2a	Given the public key $(e, n) = (11, 65)$, encrypt plain text $J B E$, where the alphabet $A, B, C, \dots X, Y, Z$ are assigned the numbers $2, 3, \dots, 26, 27$. Give the cipher text and also find the private key d .	08	4	4

- 2b By using the Euclidean algorithm, find the greatest common divisor d of 1389 and 2567 and then find integers x and y to satisfy $1389x + 2567y = d$. Also show that x and y are not unique. 08 3 3

- 3a Find a and b such that the surfaces $ax^2y + z^4 = 12$ and $5x^2 - byz = 9x$ intersect orthogonally at $(1, -1, 2)$. 06 3 3

- 3b Angle between tangents to the curve $x = t^2 + 1, y = 4t - 3, z = 2t^2 - 6t$ at $t = 1$ and $t = 2$. 05 2 2

- 3c Show that the divergence of the vector field $\phi(r)\vec{r}$ is $3\phi(r) + r\phi'(r)$, where $\vec{r}$ is the position vector of the point (x, y, z) and $r = |\vec{r}|$. 05 3 2

OR

- 4a If $\phi = x^2yz + xyz + 4xz^2$, find $\text{div}(\text{grad } \phi)$ at the point $(1, 2, 1)$. 05 1 1

- 4b Show that the field $\vec{F} = 3x^2y\hat{i} + (x^3 - 2yz^2)\hat{j} + (3z^2 - 2y^2z)\hat{k}$ is conservative vector field. Hence determine its scalar potential $\emptyset$ such that $\vec{F} = \text{grad } \emptyset$. 06 4 3

- 4c Show that $r^n\vec{r}$ is an irrotational vector for any value of n , where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $|\vec{r}| = r$. 05 3 2

- 5a If $\vec{F} = (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0, 0, 0)$ to $(1, 1, 1)$ along the curve given by $x = t, y = t^2, z = t^3$. 08 2 2

- 5b Using Stokes theorem evaluate $\int_C \vec{F} \cdot d\vec{r}$ for the vector field $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ where C is the rectangle bounded by the lines $x = -a, x = +a, y = 0, y = b$. 08 3 3

OR

- 6a Apply Divergence theorem to compute $\iint_S \vec{F} \cdot \hat{n} dS$ where $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and S is the surface of the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$. 08 3 3

- 6b If $\vec{F} = x^2\hat{i} + xy\hat{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0, 0)$ to $(1, 1)$ along the parabola $y = \sqrt{x}$. 08 2 2

- 7a Solve the differential equation: $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} - \frac{dy}{dx} - y = \cos 2x$. 08 2 2

- 7b Obtain the general solution of the second-order differential equation $x^2 y'' - xy' + y = x^2 \log(x)$. 08 2 3

OR

- 8a Solve the boundary value problem: $\frac{d^2y}{dx^2} + 4y = \sin(2x), y(0) = 1, y\left(\frac{\pi}{2}\right) = 0$. 08 4 4

- 8b Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = e^{-x} \log x$ by the method of variation of parameters. 08 3 3

- 9a The population of a town is given by the following table: 08 3 3

Year	1951	1961	1971	1981	1991
Population in thousands	19.96	39.65	58.81	77.21	94.61

Using suitable interpolation formula, calculate the increase in the population from the year 1955 to 1985.

The following table gives the viscosity of an oil as a function of temperature. Use suitable interpolation formula to compute viscosity of oil at the temperatures of 120°C and 140°C .

9b

Temperature	110	130	160	190
Viscosity	10.8	8.1	5.5	4.8

08 3 2

OR

The following table shows the observed temperatures T in degree centigrade of cooling water in a vessel at different times t (in minutes).

10a

t	1	3	5	7	9
T	85.3	74.5	67.0	60.5	54.3

8 3 3

(i) Estimate the temperature at $t = 1.5$.

(ii) Estimate the approximate rate of cooling at $t = 3$.

The following data defines the sea-level concentration of dissolved oxygen for fresh water as a function of temperature:

10b

$T(^{\circ}\text{C})$	0	8	16	24	32
$O_2(\text{mg/L})$	14.621	11.843	9.870	8.418	7.305

08 3 2

Use appropriate Newton's interpolation formula to calculate the amount of oxygen when temperature is 10°C and 35°C .

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

II Semester B. E. Regular / Supplementary Examinations Aug-2024

(Common to AI, BT, CS, CY, CD & IS)

NUMBER THEORY, VECTOR CALCULUS AND COMPUTATIONAL METHODS

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Mathematics hand book to be provided.

PART-A

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1	1.1	The sum of positive divisors of 864 is _____.	02	1	1										
	1.2	Compute 6^{100} modulo 7.	02	2	2										
	1.3	If $\phi(x, y, z) = xy^2z^3 - x^3y^2z$ then find $\nabla\phi$.	02	2	2										
	1.4	Show that the vector function $\vec{f} = 2xyz\hat{i} + (xy - y^2z)\hat{j} + (x^2 - zx)\hat{k}$ is solenoidal.	02	1	2										
	1.5	If $\vec{F}$ represents the velocity of a fluid then $\int_C \vec{F} \cdot d\vec{r}$ represents _____ of $\vec{f}$ around C and if $\vec{F}$ is force then $\int_C \vec{F} \cdot d\vec{r}$ represents _____.	02	1	1										
	1.6	The value of $\int_C (y^2 + 2yx)dx + (2xy + x^2)dy$ where C is the circle $x^2 + y^2 = 9$ is _____.	02	1	1										
	1.7	Solve the differential equation $\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 0$.	02	1	2										
	1.8	Find the particular integral of $(D^2 + 4)y = 2 \sin x \cos x$.	02	2	2										
	1.9	Construct the forward difference table: <table border="1"><tr><td>x</td><td>0</td><td>2</td><td>4</td><td>6</td></tr><tr><td>y</td><td>-1</td><td>3</td><td>8</td><td>11</td></tr></table>	x	0	2	4	6	y	-1	3	8	11	02	1	1
x	0	2	4	6											
y	-1	3	8	11											
	1.10	Find x at $y = 2$ given: <table border="1"><tr><td>x</td><td>0</td><td>1</td><td>3</td></tr><tr><td>y</td><td>1</td><td>3</td><td>4</td></tr></table>	x	0	1	3	y	1	3	4	02	2	2		
x	0	1	3												
y	1	3	4												

PART-B

2	a	By using the Euclidean algorithm, find the greatest common divisor d of 1769 and 2378 and then find integers x and y to satisfy $1769x + 2378y = d$.	08	3	3
	b	Find the last two digits of 3333^{4444} .	08	3	3
3	a	A particle moves along the curve $\vec{r} = 2t^2\hat{i} + (t^2 - 4t)\hat{j} + (3t - 5)\hat{k}$. Find the components of velocity and acceleration in the direction of the vector $\vec{c} = \hat{i} - 3\hat{j} + 2\hat{k}$ at $t = 1$.	08	2	2
	b	Find the constants a and b so that the surface $3x^2 - 2y^2 - 3z^2 + 8 = 0$ is orthogonal to the surface $ax^2 + y^2 = bz$ at the point $(-1, 2, 1)$.	08	3	2

OR

4	a	Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = \vec{r} $, then prove that $\nabla^2 f(r) = f''(r) + \frac{2}{r}f'(r)$.	08	2	2														
	b	Show that $\vec{f} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$ is irrotational. Find the function ϕ such that $\vec{f} = \text{grad } \phi$.	08	3	2														
5	a	Find the work done in moving a particle in the force field $\vec{f} = 3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}$ along the straight line $(0, 0, 0)$ to $(2, 1, 3)$.	08	3	3														
	b	Show that area enclosed by a simple closed curve C is given by $\frac{1}{2}\oint_C \{x dy - y dx\}$. Using this, find the area bounded by the ellipse with axes $2a$ and $2b$.	08	3	4														
OR																			
6	a	Using Divergence theorem, evaluate $\iint_S [(x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}] \cdot \hat{n} ds$ over the surface of the rectangular parallelepiped $0 \leq x \leq a$, $0 \leq y \leq b$, $0 \leq z \leq c$.	08	3	3														
	b	Evaluate by Stokes theorem $\oint_C (x + y)dx + (2x - z)dy + (y + z)dz$, C is the boundary of the triangular surface with vertices $(0, 0, 0)$, $(1, 0, 0)$, and $(1, 1, 0)$.	08	3															
7	a	Solve the differential equation: $(D^2 - 4D + 4)y = x^2e^{2x} + \sin^2 x$.	08	2	3														
	b	Solve the differential equation $y'' + a^2y = \sec(ax)$ using the method of variation of parameters.	08	3	3														
OR																			
8	a	Find the general solution of the differential equation: $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = (x^2 + 7x + 5)$	08	2	3														
	b	Solve $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = \log x$	08	3	3														
9	a	The table gives the distance in nautical miles (y) of the visible horizon for the given heights (x) in feet above the earth's surface: <table><tr><td>x</td><td>150</td><td>200</td><td>250</td><td>300</td><td>350</td><td>400</td></tr><tr><td>y</td><td>13.03</td><td>15.04</td><td>16.81</td><td>18.42</td><td>19.90</td><td>21.27</td></tr></table>	x	150	200	250	300	350	400	y	13.03	15.04	16.81	18.42	19.90	21.27			
x	150	200	250	300	350	400													
y	13.03	15.04	16.81	18.42	19.90	21.27													
	b	Find the values of y when $x = 160\text{ft}$ and $x = 410\text{ft}$. Find y' and y'' at $x = 1.2$, given: <table><tr><td>x</td><td>1.0</td><td>1.2</td><td>1.4</td><td>1.6</td><td>1.8</td><td>2.0</td></tr><tr><td>y</td><td>2.72</td><td>3.32</td><td>4.06</td><td>4.96</td><td>6.05</td><td>7.39</td></tr></table>	x	1.0	1.2	1.4	1.6	1.8	2.0	y	2.72	3.32	4.06	4.96	6.05	7.39	08	3	4
x	1.0	1.2	1.4	1.6	1.8	2.0													
y	2.72	3.32	4.06	4.96	6.05	7.39													
OR																			
10	a	By using the Lagrange's interpolation formula fit a polynomial to the data given: <table><tr><td>x</td><td>0</td><td>1</td><td>3</td><td>4</td></tr><tr><td>y</td><td>-12</td><td>0</td><td>6</td><td>12</td></tr></table>	x	0	1	3	4	y	-12	0	6	12							
x	0	1	3	4															
y	-12	0	6	12															
	b	Hence find y when $x = 2$. The following table gives the temperature θ (in degree Celcius) of a cooling body at different instants of time t (in seconds) <table><tr><td>t</td><td>1</td><td>3</td><td>5</td><td>7</td><td>9</td></tr><tr><td>θ</td><td>85.3</td><td>74.5</td><td>67.0</td><td>60.5</td><td>54.3</td></tr></table>	t	1	3	5	7	9	θ	85.3	74.5	67.0	60.5	54.3	08	3	4		
t	1	3	5	7	9														
θ	85.3	74.5	67.0	60.5	54.3														
Find approximately the rate of cooling at $t = 3\text{ seconds}$ and $t = 7\text{ seconds}$.																			
			08	2	3														